

EECS 352 Group Final Project

Project Proposal

You must turn in a project proposal between 500 and 1000 words

The submission must be an ADOBE ACROBAT PDF FILE.

You must clearly and directly answer the boldfaced questions below. Feel free to delete the text in italics.

- **What is the Project title?**

Uke Can Do It!

- **Who is in the group? Who is doing this for credit? A non-for-credit can only be a 3rd group member. So if you're not for credit, but want to contribute, find a group of 2.**

Give full names and netIDs

Rohan Sekhar ljn1139 – For credit

Moksh Srinivas mss1672 – For credit

- **Give a high-level description of what you want to do. Are you solving a problem? Duplicating a system you think is cool? Running an experiment? (a couple of sentences)**

We want to create an interactive game to help people learn how to play the ukulele. Similar to the video game Just Dance where hitting the right dance moves earns you points, we want to create a game that tells the user which chords to play for their favorite songs, and gives feedback via score for how accurately and on time chords were played.

- **Explain why this is this an interesting/useful/cool thing to do. (a paragraph)**

This is a cool project idea because it takes something that some people find mundane (practicing their instrument) and gamifies it in a way that adds fun to the learning process. It also gives people a tangible way to evaluate their practice quality which normally would just rely on human judgment.

- **What prior art is there? Give citations. Is there a commercial product that does something like this? Describe it and give links? Include a minimum of 4 relevant research papers or example projects. I want full bibliographic information.....AND hyperlinks to research papers related to the project. READ a couple of them before submitting the proposal. Try scholar.google.com as a starting point. (a paragraph or two)**

Yousician (<https://yousician.com/>): Listens to the player through the device microphone, scrolls upcoming notes and chords across the screen (like Guitar Hero), and gives real-time feedback scored on accuracy and timing.

Rocksmith (<https://www.ubisoft.com/en-us/game/rocksmith/plus>): Either receives input through connecting guitar to computer using a usb or accepts computer microphone for acoustic instruments and analyzes incoming signal in real time.

Simply (<https://www.hellosimply.com/>): Tablet/phone app, works across multiple instruments. You play the instrument and the device takes the mic input and determines if the correct key is played.

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Our system is using the Librosa library ([McFee, Brian, et al. "librosa: Audio and music signal analysis in python." SciPy 2015.18-24 \(2015\): 7.](#)) for the scoring system. We will be using onset detection, tempo deviations, and chroma functions for chord recognition and timing analysis.

- **What is the EXACT TASK that your system will do? Described the exact user interaction you expect (a paragraph or two).**

The user will select an available song chord progression along with a target tempo. The screen displays the chords in sequence like Guitar Hero , prompting the user to strum each chord at the correct time. The user will play along on a real ukulele while the system records their performance through the computer microphone.

Afterward, the system analyzes the recording and scores it on chord correctness, timing, and how cleanly each chord was played. The output shows an overall score, a chord-by-chord breakdown, timing accuracy, and a visual timeline of mistakes.

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- **What measure will you use to evaluate performance of what you build? Be specific. Don't say just "We'll measure performance."**

To measure performance, we will create test recordings that are controlled for certain performance features (good bad accuracy, on off tempo, muted chords) with predetermined objective scores, and then measure how close our program replicates those results.

- **Is there a data set that your system can be tried on? If so, what is it (give links). If not, explain why not. If there is no existing data set, where will you get it? Do you have to label it yourself or is it already labeled? How big is it? How many examples? How is it encoded?**

Our system doesn't require a large training dataset because we aren't training an ML model. Instead, we use audio processing functions to evaluate each recording directly, computing features like onset times and chroma vectors, comparing those features against the expected chord progressions and tempo.

We do need a small test dataset to evaluate whether our system works. Since there's no off-the-shelf dataset that matches our task of graded beginner ukulele performances, we'll record our own samples: around 20 short ukulele recordings of simple chord progressions including correct and intentionally incorrect performances.

We'll label these ourselves, with the expected chord, strum time, and known mistake type for each chord event. This data is used only to measure how accurately our scoring functions identify audio features, not to train the system.

- **What is the baseline approach you will compare your system to? Random performance? An existing system? Human performance? Be specific.**

Our baseline approach is an onset-only scorer. It will detect whether a strum happened near each expected chord time but won't identify which chord was played so any on-time strum is deemed correct. This gives a simple comparison point for timing accuracy without using chroma based chord recognition.

We compare our full system against this baseline on the same set of recorded ukulele examples. The baseline handles detecting missed or late strums, but should fail when the user strums on time but plays the wrong chord. Our system improves on the baseline using chroma features to distinguish between chords, allowing it to catch wrong chord mistakes that the baseline would incorrectly mark as correct.

- **Describe any existing software packages you will use. Where will you get them? Give links. Have you already tried them out? If not, why not? If so, how did that go**

- **Librosa:** <https://librosa.org/>
Used for audio analysis, including loading recordings, detecting strum onsets, estimating tempo, and computing chroma features for chord recognition.

MediaRecorder API: <https://developer.mozilla.org/en-US/docs/Web/API/MediaRecorder>
Used to record the user's ukulele performance directly through the browser microphone.

Next.js: <https://nextjs.org/docs>
Used to build the game-style web interface where chords appear on screen and the user can start/stop recording.

FastAPI: <https://fastapi.tiangolo.com/>
Used to connect the frontend to our Python audio-analysis code. The frontend would send the recorded audio file to the backend, and the backend would return the score breakdown.

NumPy: <https://numpy.org/>

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Used for numerical calculations, such as comparing detected chroma vectors to expected chord templates.

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- **What are potential obstacles to success?**

Do you need to learn a new language?

Is there some data you don't know how to collect?

Something else?

One potential obstacle is that the chord recognition from microphone audio could be noisy unreliable if there is background noise or messy chord transitions. It also might be difficult to determine time alignment since the system needs to decide whether the user strummed early on time late.

- **How will you QUICKLY determine if these obstacles will stop you? Describe a CONCRETE way of finding out if these obstacles will make it impossible to complete your plan in the next month. You want to know early, so you can change your plan.**

Instead of building out the game interface first, we'll build out the chord timing detection software first because the feasibility of that portion determines where we pivot.

- **What other tasks (besides coding) will you need to do?**

Data collection?

User survey?

What do you have to learn/know/get before you can start these tasks?

We will need to collect and label our own small evaluation dataset. This means recording short ukulele performances of fixed chord progressions, including both correct examples and intentionally incorrect examples. We will need examples with wrong chords, missed strums, late strums, early strums, and unclear messy chord transitions.

- **Milestones**

What will you complete by the first project meeting?

What will you complete by the second project meeting?

What will you complete by the final presentation?

Who is responsible for each milestone?

First project meeting: we'll complete a basic proof of concept which involves recording a few ukulele audio clips, loading them into python, computing chroma features, detecting strum onsets, and testing simple chord template matching on a small set of chords.

Second project meeting: we'll complete the main scoring pipeline and small evaluation dataset. The system should take a recorded performance, compare it to a known chord progression and tempo, and output chord accuracy, timing accuracy, and a basic overall score. We'll also begin building the web interface.

Final presentation: we'll complete the full demo system, project website, and final evaluation. A

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user can play through a short ukulele progression, record the attempt, receive a score breakdown, and view a timeline of correct chords, wrong chords, missed strums, and timing mistakes.